Task 8: LangChain and RAG

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Goal

1. Reimplement LINC using LangChain
2. Use RAG to query the knowledge base to give models additional context
3. Use LangChain with logical inference

Approach

This is LINC's original pipeline: ProofWriter → Semantic Parser (LLM) → FOL → Prover9 → results (label vs. prediction).
This is my pipeline (LINC+): Prolog → LLM-generated ProofWriter → RAG → Semantic Parser (LLM) → Prover9 → Prover9 trace and results.

→ To view the queries/conclusions and their trace, check out "query_results.txt":

LangChain and LINC

I took my time learning LangChain and its libraries. I primarily used LangChain to chain the prompt, LLM, and output. This chain really simplified a lot of code. I am a big fan of LangChain and will be using it for a while. I enjoyed reimplementing LINC using it.

Using RAG and Querying

For this part of the task, I decided to use FAISS. I spent some time understanding how FAISS works. It is somewhat similar to k-nearest neighbors. I loaded the generated ProofWriter knowledge base into the vector store and then used a nearest-neighbor search to find similar ProofWriter examples. Afterwards, I gave the semantic parser LLM similar ProofWriter examples to show it the general structure of ProofWriter examples and to expand its domain.

Using Langchain With Logical Inference

I decided to use LangChain to help explain logical inference. Prover9 uses neither backwards nor forward chaining. Prover9 uses resolution-based inference. Prover9 does a proof by contradiction. To get an inference trace, I used nltk's wrapper _call_prover9(). The wrapper captures an stdout trace of Prover9's inference. Afterwards, for readability, I used LangChain to invoke an LLM chain that translates the trace into natural language.

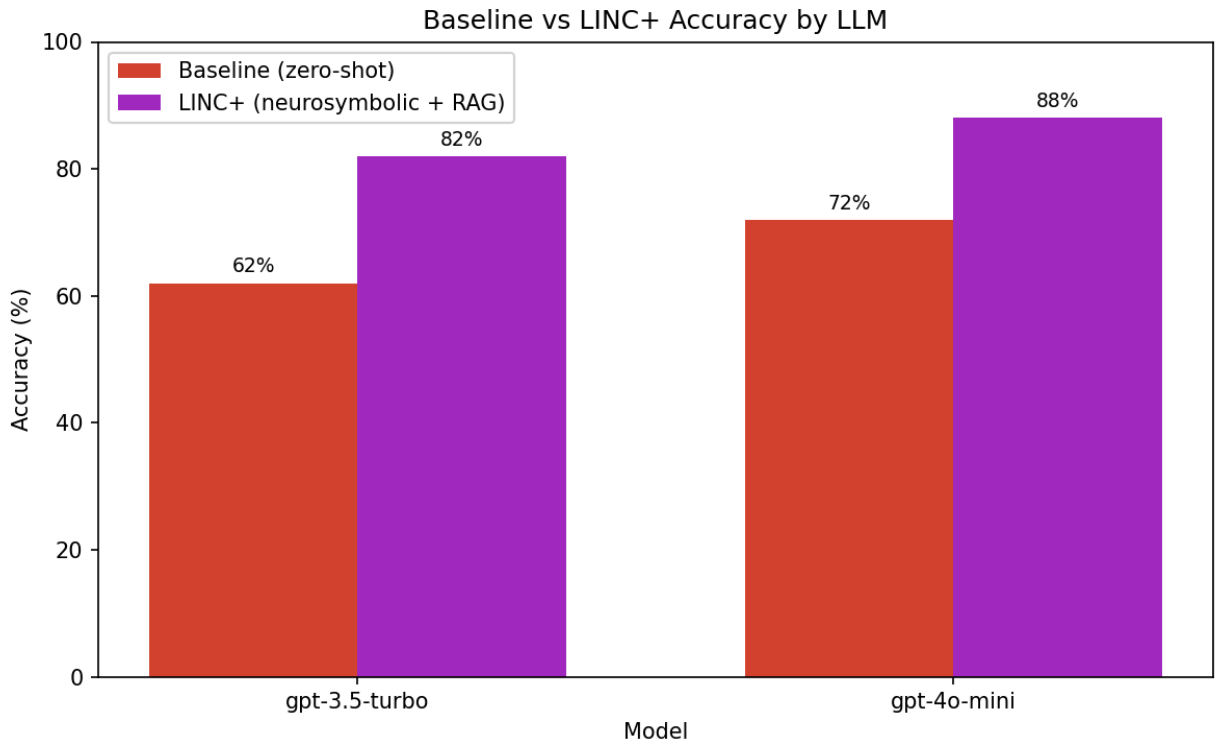
Further Improvements

I have an idea for a better use of RAG that might deliver better results than the LINC paper. I will test it out in task 9. LINC shows the semantic parser LLM a random set of FOLIO (an NL and FOL dataset) examples to help it translate NL to FOL. I would like to use RAG to show the semantic parser LLM similar FOLIO examples to the example it's currently facing. I will translate my KB into ProofWriter and FOLIO examples to start with.

Results

Querying/testing conclusions: please find “results.txt.”

Accuracy (results.png) (not really discussed too much above):
I will use RAG on my KB’s FOLIO samples in task 9, then compare it with this graph that uses ProofWriter examples. We should see much better results.



References

Papers

LINC's study: https://arxiv.org/abs/2310.15164
ProofWriter: https://arxiv.org/abs/2012.13048

AI-Assisted Coding

Used Claude to:

- Better understand LINC's preexisting code in more detail
- Searching through library functions (e.g., from NLTK, Prover9, and LangChain)
- Assist with debugging where appropriate

I critically evaluated all outputs. I wrote all the code or got it from LINC and made changes. I also attributed LINC where appropriate.

→ To understand the code better, refer to the README.md file.